1 A programmer intends to code the backend of an e-commerce platform using Object-Oriented Programming (OOP). The platform will collect the user id, password, email addresses and mobile numbers of all customers. There are methods to update the shopping carts and calculate total cost of an order.

The platform also has a 2 membership schemes. Platinum members will have their shipping fees for an order waived, if the total cost of an order exceeds $200. Prestige members pay $20 to purchase $100 worth of cash rebates. The rebates can be used to offset their purchases if terms and conditions are met.

(a) Draw a UML class diagram to show the following for the situation described above:

- the superclass
- any subclasses
- inheritance
- properties
- appropriate methods. [6]

(b) State the purpose of inheritance in OOP and explain how it is achieved, using the example above. [3]

(c) State **one** situation that could threaten the privacy of customer data that the platform has collected and suggest **one** measure the e-commerce platform should take to protect the data that it has collected. [2]

(d) As e-commerce becomes more and more popular, describe **one** social impact of this trend. [2]

2 An art education company has engaged you to develop a membership web portal. The portal should enable users to register to be a member, and current members to login to access other portal e-services.

(a) State **one** usability principle and describe how the user interface of the registration and login module of the membership portal should be designed to demonstrate this principle. [2]

(b) As your team works on the login module of the membership portal, you realised that your team has failed to incorporate appropriate data or input validation checks as well as input verification.

Explain **one** input validation check that you would like to your team to incorporate in the login module. [1]

3 In computer science, a priority queue is an abstract data type which is similar to a queue data structure in which each element “additionally” has a priority associated with it. An element of higher priority is served first and if two elements have the same priority, they are served according to the order that they are enqueued.

One way to implement a priority queue is to use a max heap. A max heap is a complete binary tree in which the priority of every internal node is at least that of its children node. When the highest priority member is served, we remove the root of the binary tree. When implementing the max heap from an array, if the element at index `i` is the parent node, the elements at indices `(2i + 1)` and `(2i + 2)` are the left and right child respectively. (See Figure 1 below)

Priority	8	6	2	2	5
Index	0	1	2	3	4

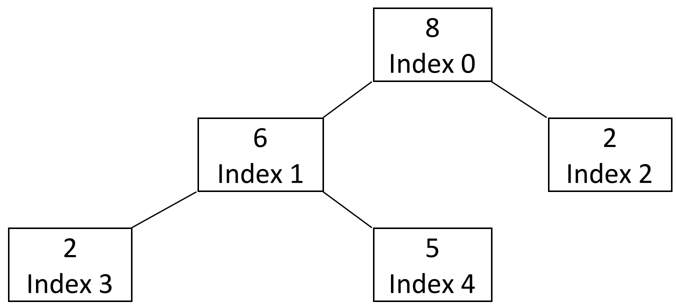

Figure 1

It is proposed that an `insert()` method, described as follows, is used to insert new elements into a priority queue. When a new element is inserted in the array, its priority is compared to that of its parent node in the max heap. If the priority of the new element is higher than of its parent node, it is swapped with its parent node. This is done recursively until the priority of the new element is not higher than of its parent node.

(a) Describe how inserting a new element is done in a sorted linked list, and state **one** possible advantage the above `insert()` method has over a sorted linked list. [3]

(b) Elements with priorities 10 and 7 are inserted into the priority queue in Figure 1. Sketch the final max heap. [3]

(c) State **one** problem the suggested algorithm for `insert()` may cause. [1]

4 A brick-and-mortar company selling household items has contacted you to discuss on the possibility of developing an application to keep track of its inventory. The company has a warehouse situated at Loyang which serves three stores across Singapore.

Currently, the company engages its employees to keep track of its store inventory manually via the store's inventory book and through phone calls to the warehouse. The company wishes to have an electronic version of its inventory book that will cut down man-hours through automation as well as to minimise mistakes due to human errors.

After the phone conversation, you call for a meeting with your team of developers to discuss whether it is beneficial for the company to develop a native application or a web-based application for the above purpose.

Describe your conversation with your team in deciding between a native application and a web-based application, by comparing **three** differences, in the context of the company's requirements. [3]

5 Table 1 below shows a part of a Civics Group (CG) 22S90 list of a Junior College, where each student must have **only one** CCA.

Civics Group: 22S90					
Civics Tutor: Mr. Von Neumann					
Index No.	**Student Name**	**CCA**	**CCA Teacher**	**Scholarship**	**Student Contact Number**
1	Charles Babbage	Science Society	Mr. Lorax	MOE-JC	87325365
2	Ben Goh	Dance	Ms. Doe	None	80764730
3	Alan Turing	Robotics Club	Ms. Am	International	87615635
4	Jane Doe	Infocomm Club	Mr. Ben	None	86720367
5	Khwarizmi	Math Society	Ms. Claire	DSTA	87453452
6	Rakesh	Dance	Ms. Doe	MOE-JC	98046782

Table 1

(a) Using the database tables `CG`, `SCHOLARSHIP` and `STUDENT`, present the data above in second normal form. Present the data as table descriptions. Explain why the resulting schema is not in third normal form. [4]

(b) Hence, present your table descriptions in **(a)** in the third normal form. [2]

(c) Draw an ER diagram for a normalised database design. [3]

(d) By stating examples in the above context, explain the significance of the following terms:

(i) primary key [2]

(ii) foreign key [2]

(e) State a scenario where a NoSQL database would be more appropriate for the above context. [1]